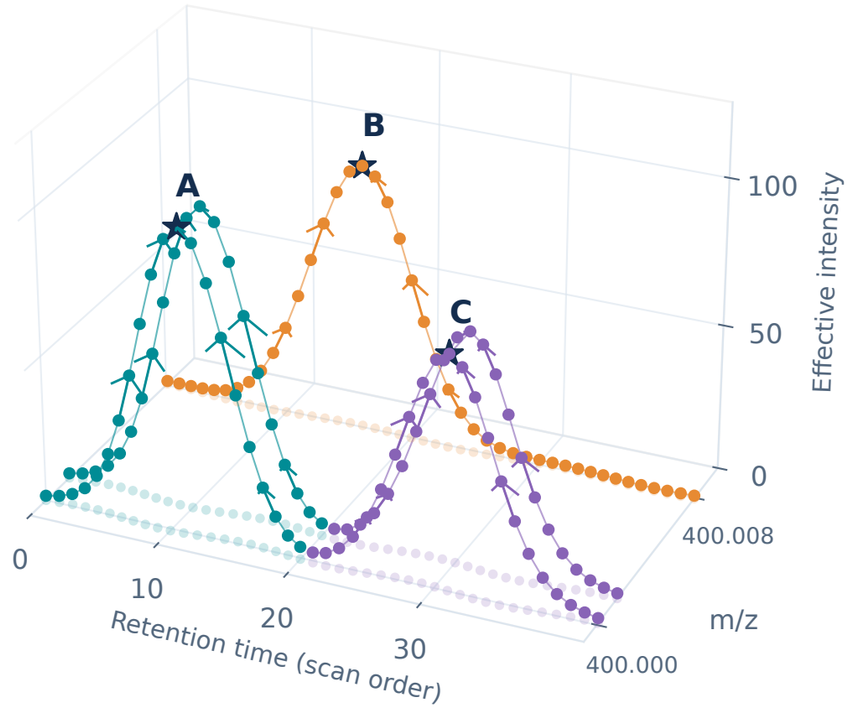
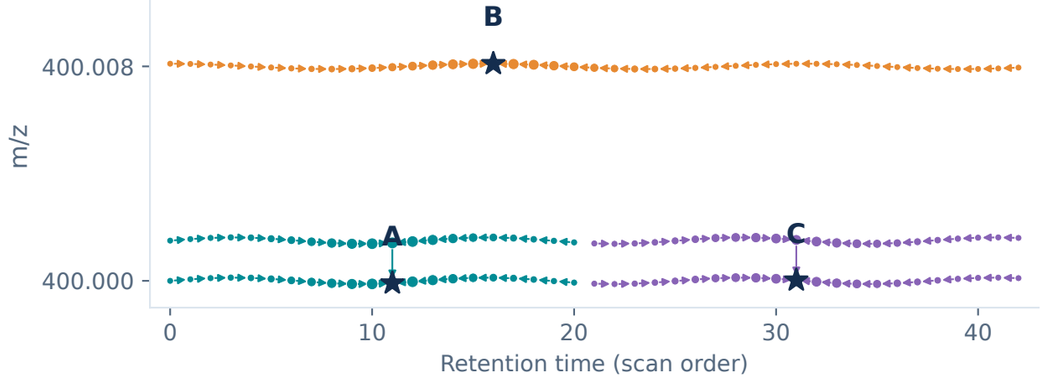
What does the 3D in Peak3D mean?

An intensity landscape over retention time and m/z, represented by connected centroids.



A and C: same m/z, different RT. B: a co-eluting ion at a different m/z.

The same graph, viewed from above



Position: retention time + m/z

Links use nearby scans and pair-specific mass tolerance.

Height: effective intensity

Same-scan split pieces contribute to the local height.

Colour: centroids assigned to the same surviving apex.

Time and m/z define local neighbours; intensity directs their paths toward apexes.

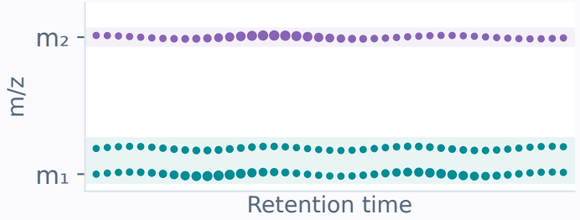
Synthetic centroid graph • actual Peak3D links and basin labels • no fitted surface or voxel grid

The difference is how peak candidates are formed

Both approaches start with retention time, m/z and intensity. They organise the data differently.

TRACE-BASED DETECTION

1 Assemble traces / ROIs



2 Find peaks within traces



3 Report peak candidates

A trace is the parent object. Chromatographic segmentation defines peaks inside it.

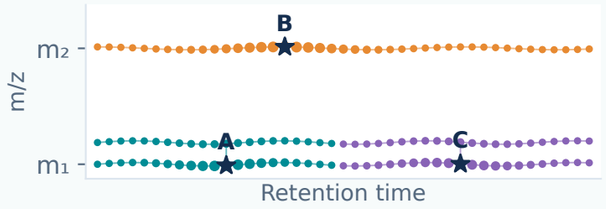
Trace building may use intensity and adaptive tolerances; implementations differ.

Peak3D CANDIDATE DETECTION

1 Link local centroids uphill



2 Form and merge apex basins



3 Validate peak candidates

A basin is the candidate object. Apply basin gates, an EIC check, and the final score.

Dot size shows intensity. Peak3D panels show actual graph and basin assignments.

Trace-based: form traces, then find peaks. Peak3D: form apex basins directly from centroids.

Architecture schematic • centWave / MassCube: ROI then segmentation; asari: composite mass tracks • sources in notes